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PROJECT REPORT

Privacy Preserving of Distributed Satellite's Attitude Control



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ABSTRACT

Many control systems contain privacy-sensitive information that could put the entire system at risk when control actions are computed on third-party platforms. With the development of embedded and decentralized control systems in industrial automation, the ability to process massive data has been provided to control aerospace systems. In this paper, a privacy-preserving algorithm is addressed for distributed computing of the attitude control of a satellite. A compensator is designed by the pole placement method and discretized for digital control implementation of the dynamic system. The feedback sensor measurements are quantized by a desired computational resolution and the computations are evaluated within the enciphering A/D and deciphering D/A converters enabling computations by using finite field arithmetic of the privacy-preserving protocols. The secure multiparty computation based on Shamir's secret sharing scheme is utilized as an approach to develop secure protocols that ensure the privacy and accuracy of subsystems and protocols.

Keywords— digital privacy, secure multiparty computation, secret sharing, networked control system, distributed control, cyber-physical system.

CHAPTER 1. INTRODUCTION

The emergence of the Internet of Things (IoT) and fifth-generation (5g) networks enable connecting different devices and exchanging data through cloud computing. However, there may be internal and external threats in these public networks. In cyber-physical (CPS) systems, data will be interpreted through predictive analytic procedures implemented in the cyber layer to calculate the estimation of the optimal states for the physical system. For example, smart buildings, smart vehicles, and smart airplanes are some of the CPS systems [1-5].

The ongoing problem of connecting CPS systems by cloud servers also enhances the demand and tendency in the field of networked control systems [6]. The operations are distributed at different geographical locations and coordinated by information exchange over communication networks [7]. Cloud control systems are employed to solve large-scale problems such as the smart grid of hydraulic systems. Although the data manipulated in these smart systems may include private information about the system owner, the available approaches are not guaranteed to protect the privacy of system information under control. The data disclosed to dishonest third-party platforms may eventually promote the risk of cyberattacks e.g., eavesdropping and sabotage actions. Hence, the upcoming control infrastructures should be reinforced against these threats.

A general architecture of CPS is depicted in figure 1-1. The computation at the top layer could interface with the physical system at the bottom layer. The cyber layer includes cloud servers, smart devices, software, and hardware. On the other hand, the physical layer consists of smart networks, vehicles, etc., at the top of the figure 1-1. The mentioned layers are integrated by a communication network that may be provided by the communication satellites. Privacy challenges may arise when either the computation is corrupted by false sensor information or the system information is disclosed to dishonest servers, sending malicious control actions to the system.

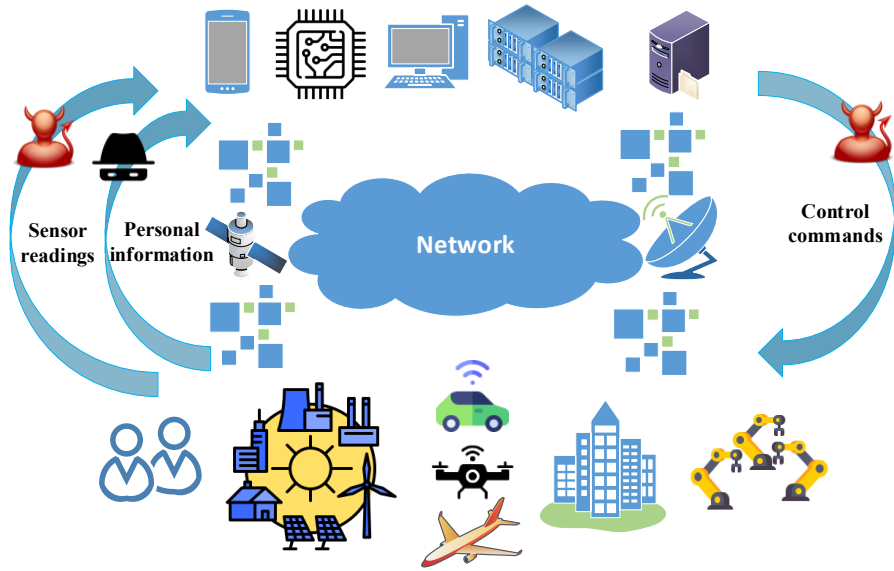


Figure 1-1 The general representation of a CPS

For example, consider a smart water grid system that provides real-time consumption data through the use of a smart water meter that collects the information of consumers [8, 9]. Another example could be intelligent transportation systems (ITS) which may be controlled by cloud computing in the future in order to make a unified air vehicular network as an example. ITS needs data from individuals containing privacy-sensitive information such as passenger origin, live location, and destination [10].



Figure 1-2 A futuristic air transportation system

A conceptual scheme of an urban air mobility environment is illustrated in figure 1-2 where air vehicles with a variety of missions and with or without pilots, are able to interact safely and efficiently [11]. Despite the fact that these smart-making projects can reduce the consumption of resources, waste of time, etc., they disclose privacy-sensitive information of the data owners.

Related works. Although the topic of cryptography and privacy is a broad field, the dynamic and control issues of the mentioned cyber-physical systems are less addressed but are attracting more attentions. Most recent studies are based on homomorphic encryption [12-15]. There are also several works that have faced with the mentioned problem by the application of differential privacy or secure communication channels, for example [16] and [17]. Though these methods are advantageous, they should rely on trusted third-party, and data owners are needed to expose their raw data to probably corrupted servers which are susceptible to adversarial attacks. Privacy against the denial-of-service attack is investigated in [18] using the design of network transmission times. The privacy preservation recursive least-squares solutions and privacy preservation in distributed optimization via dual decomposition and ADMM are proposed in [19] and [20], respectively by Wisniewski and colleagues, which are based on the secret sharing scheme. Privacy preserving distributed summation in a connected graph is another related work studied in [21] by the above mentioned team. Private computation of polynomials in a networked system is suggested by [22], which is based on a secret sharing scheme. They do not rely on a trusted third party as we do by our proposed method. In addition, there are some differences in privacy preserving implementation protocols. Moreover, an actual physical system is not investigated by those researchers to observe the exact performance of algorithms.

In this paper, a novel privacy preserving distributed control method for satellite's attitude is proposed where it is required to preserve the privacy of estimated and controlled parameters and signals. Primarily, equations of motion (EOMs) for the 2-DOF plant are extracted and damping of the system is obtained using modal analysis. Afterward, a compensator including an estimator and a controller is designed based on control objectives and discretized into the digital compensator. The control gains are calculated using the pole-placement method. Secure multiparty computation (SMPC) based on Shamir's secret sharing scheme (Shamir's SSS) is utilized to state distributed privacy preserving satellite's attitude control algorithm. To demonstrate the robustness of the designed digital secure multiparty control algorithm, it is assumed that the communication data of the agents are noisy.

CHAPTER 2. PROBLEM STATEMENT

A space program example is taken from [23], where it is required to control the pointing direction of a satellite in orbit about the earth. A schematic of a geosynchronous communications satellite is observed in figure 2-1. The vehicle is considered to have an astronomical survey mission requiring accurate pointing of a scientific sensor package. This package must be maintained in the quietest environment possible, which entails isolating it from the vibrations and electrical noise of the main service body and from its power supplies, thrusters, and communication gear. The resulting unrestrained non-collocated active structure is modeled as two rotary masses connected by a flexible torsional shaft.

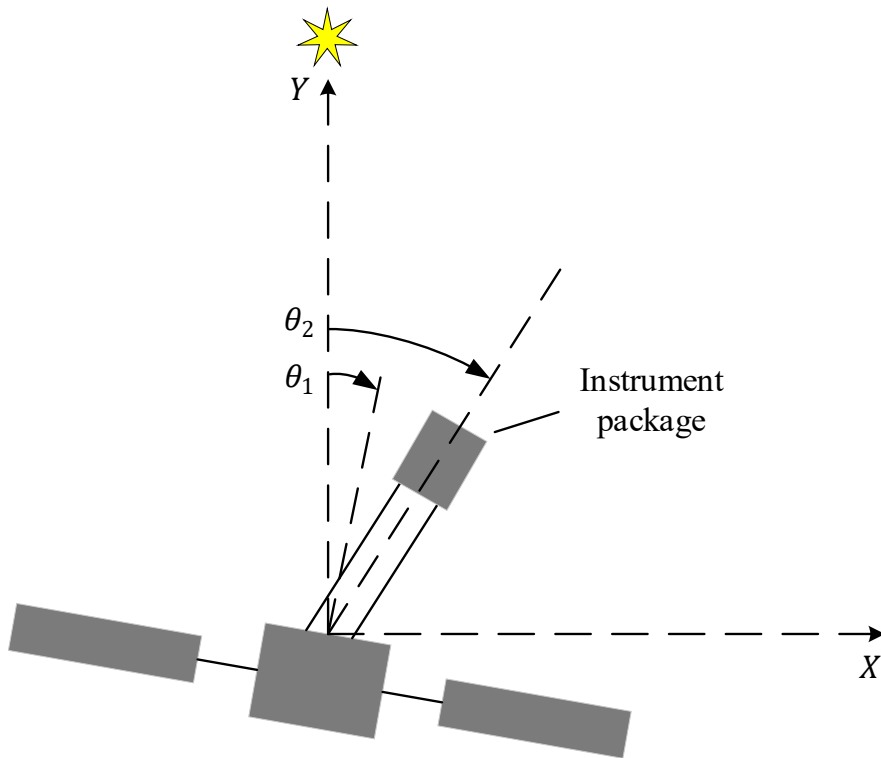


Figure 2-1 Diagram of a satellite

2.1. MATHEMATICAL MODEL

Assume two rotary masses connected by a torsional spring with an external constant torque and viscous damping, for which the equivalent mechanical system of satellite is illustrated in figure 2-2 and the linear model equations of motion may be formulated as

$$J\ddot{\theta} + B\dot{\theta} + K\theta = M \quad (2.1)$$

where $\theta = [\theta_1 \ \theta_2]^T$ is the vector of the angles of the main satellite and instrument package with respect to a very far star (the inertial reference frame), $\dot{\theta} = [\dot{\theta}_1 \ \dot{\theta}_2]^T$, and $\ddot{\theta} = [\ddot{\theta}_1 \ \ddot{\theta}_2]^T$ are the vectors of angular velocity and acceleration, respectively. $M = [T_c \ 0]^T$ is the vector of applied torques to the vehicle, where T_c is the control torque. J , B , and K are the mass moment of inertia, torsional damping, and torsional stiffness matrices specified as, $J_{11} = J_1$, $J_{12} = J_{21} = 0$, $J_{22} = J_2$, $K_{11} = k$, $K_{12} = K_{21} = -k$, $K_{22} = k$, $B_{11} = c$, $B_{12} = B_{21} = -c$, $B_{22} = c$.

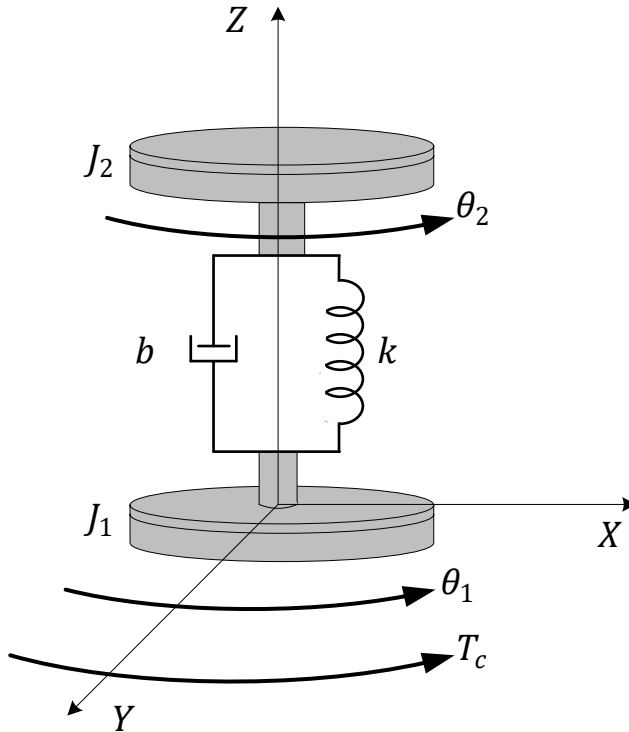


Figure 2-2 two-body model of the satellite

A state space representation of the system is,

$$\dot{x} = Fx + Gu \quad (2.2)$$

where $x = [\theta_2 \ \dot{\theta}_2 \ \theta_1 \ \dot{\theta}_1]^T$ is the continuous-time state variables vector of the satellite and $u = T_c$. The state space matrices for the system are obtained as,

$$F = \begin{bmatrix} 0 & 1 & 0 & 0 \\ -\frac{k}{J_2} & -\frac{b}{J_2} & \frac{k}{J_2} & \frac{b}{J_2} \\ 0 & 0 & 0 & 1 \\ \frac{k}{J_1} & \frac{b}{J_1} & -\frac{k}{J_1} & -\frac{b}{J_1} \end{bmatrix}, G = \begin{bmatrix} 0 \\ 0 \\ 0 \\ \frac{1}{J_1} \end{bmatrix}, \quad (2.3)$$

Given system sensor mounted to the disk associated with θ_2 , the observation equation is

$$y = Hx + Du \quad (2.4)$$

where

$$H = [1 \ 0 \ 0 \ 0], D = 0. \quad (2.5)$$

2.2. MODAL ANALYSIS

Firstly, the natural frequencies and modes shape of the system are obtained by solving the eigenvalue problem as follows, where ω is the natural frequency of the system [24]. For this aim, mass moment of inertia and stiffness matrices are parametrized (It is assumed that $J_1 = 10j$ and $J_2 = j$).

$$J = \begin{bmatrix} 10 & 0 \\ 0 & 1 \end{bmatrix} j, K = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} k \quad (2.6)$$

$$\|K - \omega^2 J\| = \det \begin{bmatrix} k_{11} - J_1 \omega^2 & k_{12} \\ k_{12} & k_{22} - J_2 \omega^2 \end{bmatrix} = 0 \quad (2.7)$$

By substituting the system's parameters of (2.6) in (2.7) the characteristic equation becomes

$$10\lambda^4 - 11\lambda^2 = 0 \quad (2.8)$$

where $\lambda^2 = (j/k)\omega^2$. Then, natural frequencies are obtained by solving (2.8),

$$\omega_{1,2} = 0, \sqrt{\frac{11}{10}} \sqrt{\frac{k}{j}} \quad (2.9)$$

The system is *unrestrained*, which can be observed in figure 2-2 [25]. The fundamental frequency of the system leads to be zero, which corresponds to a *rigid body mode*. Another property of an unrestrained system is that $\det[K] = 0$. The mode shapes of the system are obtained as follow,

$$\|K - \omega_i^2 J\| \varphi_i = 0 \quad (2.10)$$

The modal matrix of the system is,

$$\varphi = \begin{bmatrix} 1 & 1 \\ 1 & -10 \end{bmatrix} \quad (2.11)$$

After specifying a proportional damping matrix for the system, the decoupled equations of motion are written using linear transformation $\theta = \varphi q$, where $q = [q_1 \ q_2]^T$ is the vector of the generalized coordinates.

$$J^{(m)} \ddot{q} + B^{(m)} \dot{q} + K^{(m)} q = \bar{M} \quad (2.12)$$

where, $J^{(m)}$, $B^{(m)}$, and $K^{(m)}$ are the generalized rotary mass moment of inertia, torsional damping, and torsional stiffness matrices, respectively. Furthermore, $\bar{M}$ is the vector of the generalized applied moments to the system. These are obtained as,

$$J^{(m)} = \varphi^T J \varphi = \begin{bmatrix} J_1^{(m)} & 0 \\ 0 & J_2^{(m)} \end{bmatrix} = \begin{bmatrix} 11 & 0 \\ 0 & 110 \end{bmatrix} j \quad (2.13)$$

$$B^{(m)} = \varphi^T B \varphi = \begin{bmatrix} B_1^{(m)} & 0 \\ 0 & B_2^{(m)} \end{bmatrix} = \begin{bmatrix} 2\xi_1 \omega_1 J_1^{(m)} & 0 \\ 0 & 2\xi_2 \omega_2 J_2^{(m)} \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 22\sqrt{110} \end{bmatrix} \xi_2 (kj)^{\frac{1}{2}} \quad (2.14)$$

$$K^{(m)} = \varphi^T K \varphi = \begin{bmatrix} K_1^{(m)} & 0 \\ 0 & K_2^{(m)} \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 12 \end{bmatrix} k \quad (2.15)$$

$$\bar{M} = \varphi^T M = \begin{bmatrix} 1 \\ 1 \end{bmatrix} T_c \quad (2.16)$$

Now, using equation (2.14), the damping matrix of the system is formed as,

$$B^{(m)} = \varphi^T B \varphi \rightarrow \varphi^{-T} \varphi^T B \varphi \varphi^{-1} = B = \varphi^{-T} B^{(m)} \varphi^{-1} \quad (2.17)$$

where the inverse of the modal matrix is calculated using equation (2.13),

$$J^{(m)} = \varphi^T J \varphi \rightarrow J^{(m)-1} J^{(m)} = J^{(m)-1} \varphi^T J \varphi = 1 \rightarrow \varphi^{-1} = J^{(m)-1} \varphi^T J \quad (2.18)$$

$$\varphi^{-1} = \begin{bmatrix} \frac{10}{11} & \frac{1}{11} \\ \frac{1}{11} & -\frac{1}{11} \end{bmatrix} \quad (2.19)$$

The damping matrix of the system is calculated by (2.18),

$$B = \begin{bmatrix} \frac{1}{11} & -\frac{1}{11} \\ -\frac{1}{11} & \frac{1}{11} \end{bmatrix} 2\sqrt{110}\xi_2(kj)^{\frac{1}{2}} \quad (2.20)$$

The experiments show that the parameters k and b change due to temperature fluctuations but are bounded to

$$0.09 \leq k \leq 0.4 \quad (2.21)$$

$$0.038\sqrt{\frac{k}{10}} \leq b \leq 0.2\sqrt{\frac{k}{10}} \quad (2.22)$$

In this way, the vehicle's natural frequency $\omega = \omega_2$ can vary between 1 and 2 rad/s , and the damping ratio $\xi = \xi_2$ varies between 0.02 and 0.1. The design procedure is to convert a digital design of an analog controller/observer for the nominal case, where parameters are selected as $\omega = 1 \text{ rad/s}$ and $\xi = 0.02$. The selected parameter values are $j = 0.1 \text{ [(kg.m}^2\text{)/rad]}$, $k = 0.091 \text{ [(N.m)/rad]}$, and $\xi = 0.02$. The system matrices become

$$J = \begin{bmatrix} 1 & 0 \\ 0 & 0.1 \end{bmatrix} \begin{bmatrix} \frac{kg.m^2}{rad} \end{bmatrix}, K = \begin{bmatrix} 0.091 & -0.091 \\ -0.091 & 0.091 \end{bmatrix} \begin{bmatrix} \frac{N.m}{rad} \end{bmatrix}$$

$$B = \begin{bmatrix} 0.0036 & -0.0036 \\ -0.0036 & 0.0036 \end{bmatrix} \begin{bmatrix} \frac{N.m.s}{rad} \end{bmatrix}$$

The generalized system matrices become

$$J^{(m)} = \begin{bmatrix} 1.1 & 0 \\ 0 & 11 \end{bmatrix} \begin{bmatrix} \frac{kg.m^2}{rad} \end{bmatrix}, K^{(m)} = \begin{bmatrix} 0 & 0 \\ 0 & 11.011 \end{bmatrix} \begin{bmatrix} \frac{N.m}{rad} \end{bmatrix}$$

$$B^{(m)} = \begin{bmatrix} 0 & 0 \\ 0 & 0.4402 \end{bmatrix} \begin{bmatrix} \frac{N.m.s}{rad} \end{bmatrix}$$

The state space matrices of the system are,

$$F = \begin{bmatrix} 0 & 1 & 0 & 0 \\ -0.91 & -0.036 & 0.91 & 0.036 \\ 0 & 0 & 0 & 1 \\ 0.091 & 0.0036 & -0.091 & -0.0036 \end{bmatrix}, G = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix},$$

The transfer function of the system is,

$$Q_s(s) = \frac{0.036(s+25)}{s^2(s^2+0.04s+1)} \quad (2.23)$$

2.3. THE CONTROL OBJECTIVES

It is a non-collocated system as the sensor and actuator are not located in the same place. The control of such systems is much more difficult than collocated ones. The goal is to design a controller/observer, where it is necessary to point the main body of the satellite in a desired direction, while the measuring sensor locates on another side body. The system satisfies a transient settling time of 20 sec and an overshoot of no more than 15%. The disturbance torques due to solar pressure, micrometeorites, and orbit perturbations are computed to be negligible.

The dynamic performance specifications will be satisfied if the closed-loop conjugate poles have a natural frequency of 0.5 rad/s and a closed-loop damping ratio of 0.5. It corresponds to an open-loop crossover frequency of $\omega_c \cong 0.5 \text{ rad/s}$ and a phase margin of about $PM = 50^\circ$.

$$\text{Settling time (tolerance} = 2\%) \ t_s = \frac{4}{\xi \omega_n} = \frac{4}{(0.5)(0.5)} = 16 \text{ s} \quad (2.24)$$

$$\text{Maximum overshoot } \%M_p = 100e^{\frac{-\pi\xi}{\sqrt{1-\xi^2}}} = 100e^{\frac{-\pi(0.5)}{\sqrt{1-(0.5)^2}}} \approx 16.5 \% \quad (2.25)$$

The next section represents the pole placement method as a control approach which is utilized to control the attitude of a satellite where the states of the system are estimated through an estimator (observer).

CHAPTER 3. SECURE MULTIPARTY COMPUTATION

Consider a set of n parties $\mathcal{P}_1, \dots, \mathcal{P}_n$ hold their inputs $s_1, \dots, s_n$ (party $\mathcal{P}_i$ holds secret s_i) and intend to evaluate a function $y = g(s_1, \dots, s_n)$ while they are not willing to disclose their individual information. Computing g such that correctness and privacy are achieved is referred to as computing g securely [26]. Secure multiparty computation (SMPC) is a theoretical framework that aims at developing protocols that allow a set of parties to evaluate the function g without revealing their private inputs. This problem has an evident solution in an *ideal world* where a trusted third party is found, but this assumption is not acceptable in the *real world* from the cryptography perspective, hence, the SMPC has a structurally decentralized framework. This should be held while some of the parties are corrupt (a subset of parties considered to be controlled by an adversary that gathers all information that corrupted parties see during the execution of the SMPC protocol). It is assumed that agents are honest but curious which means they follow the protocol's instructions but the adversary might try to deduce information from the messages that corrupted parties have received. SMPC utilizes cryptographic techniques in order to hide private data and can include individual or hybrid approaches between techniques such as secret sharing, oblivious transfer, garbled circuits, (threshold) homomorphic encryption, etc. The SMPC protocols proposed in this paper are based on secret sharing which is a popular and frequently used encryption technique with low computational complexity. Secret sharing allows a party (dealer) to distribute the secret between a set of parties so that only an authorized subset of parties is able to reconstruct the secret. The next definition provides a definition of a secret sharing scheme.

definition 1. (Secret Sharing Scheme) A secret sharing scheme allows a data owner (dealer) who has a secret to distribute its secret between a set of parties while these chunks separately reveal nothing about the secret. Indeed, some or all of the pieces of information are needed to reconstruct the secret. The piece of information is referred to as a *share*. More precisely, a secret sharing scheme is realized by meeting two requirements of *correctness* and *perfect privacy*. Correctness means the secret can be reconstructed by any authorized set of parties and perfect privacy means every unauthorized set cannot learn anything about the secret. However, by mitigating these requirements to correctness with high probability and non-identical probability distribution with low statistical distance in the domain of secrets, statistical secret sharing schemes can be generalized. A *t-threshold* secret sharing scheme for n parties contains two functions *Share* and *Reconstruct*. The *Share* function takes an integer secret s modulo some prime p as input and creates n shares $s_1, \dots, s_n$ also modulo p , and the *Reconstruct* function takes any authorized set of these shares (at least $t + 1$

members) to produce secret s . Perfect privacy of the secret sharing scheme assures that a t -threshold scheme is unauthorized with any set of t shares or fewer (i.e., secret cannot be reconstructed). An important fact is that, as mentioned, secret sharing schemes are frequently built based on a finite field $\mathbb{F}_p = \{0, \dots, p-1\}$ where p is a sufficient large agreed prime number in order to avoid wrap-arounds, and all operations are modulo p , although there are proposed schemes that are not based on a finite field. However, values must be quantized before creating shares of the secrets that will cause quantization errors. This will be discussed in chapter 4, where scaling is utilized to define and reduce the quantization error of analog data. Let $\mathcal{P}_1, \dots, \mathcal{P}_n$ be parties. Assume that there exists a secure channel between every pair of $(\mathcal{P}_i, \mathcal{P}_j)$ for $i, j = 1, \dots, n$ and $i \neq j$ and that $\{s_{i,j}\}$ for $j = 1, \dots, n$ are shares of s_i which belongs to party $\mathcal{P}_i$. When a party, $\mathcal{P}_i$, distributes the shares of s_i , it means that $\mathcal{P}_i$ securely sends $s_{i,1}$ to $\mathcal{P}_1$, $s_{i,2}$ to $\mathcal{P}_2$, and $s_{i,n}$ to party $\mathcal{P}_n$. when a party $\mathcal{P}_i$ broadcasts a value a , it means that $\mathcal{P}_i$ makes a known to all parties. $\llbracket s \rrbracket$ represents a set of shares of s . In a secret sharing scheme, for two secrets a and b , parties are able to compute shares of the sum $\llbracket a + b \rrbracket$ and of the product $\llbracket ab \rrbracket$ modulo p , only through local computation on corresponding shares of $\llbracket a \rrbracket$ and $\llbracket b \rrbracket$ which are a particular form of the ciphertext. To see this, consider an additive sharing scheme where the dealer draws $n-1$ shares $s_1, \dots, s_{n-1}$ uniformly at random over $\mathbb{F}_p$ and chooses s_n so that $s = s_1 + \dots + s_n$. As its name suggests, this scheme satisfies the addition of shares to be computed correctly by servers. Table 3-1 assumes $s \in \mathbb{F}_{13}$ and three agents $\{\mathcal{P}_i\}_{i \in \{1,2,3\}}$, have private values $\{2,3,5\}$ respectively and want to learn the sum of their values.

Table 3-1 An example of the additive secret sharing scheme with 3 agents

Agent	Uniform random numbers	Received shares	Shares Creation (mod 13)	$d_i = \sum_{j=1}^n s_{j,i} \text{ mod } 13$	$s = \sum_{i \in \{1,2,3\}} d_i \text{ mod } 13$
1	7.3	2 = 7 + 3 + 5	5.10.8	5 + 10 + 8 = 10	10 + 12 + 1 = 10
2	10.3	3 = 10 + 3 + 3	3.7.2	3 + 2 + 7 = 12	
3	8.2	5 = 8 + 2 + 8	8.3.3	8 + 3 + 3 = 1	

As can be observed in table 3-1, parties create shares of their secrets using the additive sharing scheme and distribute shares. Afterward, parties broadcast the sum of

received shares and now all three parties can learn the sum of the secrets by adding these new broadcast values. Multiplication is more complicated as it requires more communication between participants, nevertheless, this can be alleviated for instance through a preprocessing phase. However, a multiplicative sharing scheme can be defined, *Shamir's SSS* is implemented in this paper that holds $t < \frac{n}{2}$ for the underlying case study [27]. Secret sharing based SMPC also allows delegating the computation to the desired set of computing parties (or servers). In fact, the input-provider (data owner) can share the inputs among the servers, and these can carry the rest of the computation, operating on the secret shared values. ultimately, the data owner can reconstruct the output.

Shamir's Secret Sharing Scheme (Shamir's SSS) The underlying idea of Shamir's SSS is that it takes $t + 1$ points $(a_1, b_1), \dots, (a_{t+1}, b_{t+1})$ in $\mathbb{F}^2$ with distinct a_i 's to determine a unique polynomial $f(x)$ of degree t such that $f(a_i) = b_i$ for $i = 1, \dots, t + 1$. Shamir's SSS uses Lagrange polynomials in order to create shares of a secret $s \in \mathbb{F}_p$. A polynomial $f(x) \in \mathbb{F}_p[x]$ of degree $t < n$ is chosen in such a way that $f(0) = s$ and the rest of the coefficients are random. Shares of the secret s can be calculated as $s_1 = f(1), \dots, s_n = f(n)$ and distributed among n parties. Shamir's SSS with input $s \in \mathbb{F}_p$ can be summarized in these steps,

Input-sharing $\Omega: s \rightarrow \llbracket s \rrbracket$. The client (input-provider) selects coefficients $\{r_1, \dots, r_t\} \in \mathbb{F}_p^t$ uniformly at random and define a polynomial

$$f(x) = s + r_1x + \dots + r_tx^t \quad (3.1)$$

in $\mathbb{F}_p$. Shares are calculated as $s_1 = f(1), \dots, s_n = f(n)$, nevertheless it can be formed in other structure with non-zero integer inputs. Then shares are distributed.

Computation phase $\llbracket Y \rrbracket: \llbracket s \rrbracket \rightarrow \llbracket sg \rrbracket$. It may contain three gates: addition, multiply-by-constant, and multiplication gates. All gates must be processed before the next stage by repeating the mentioned operations. The computation phase of the Shamir's SSS is completely stated in [26]. g represents the SMPC function performed on received shares of the secrets.

Output reconstruction $\mathcal{R}: \llbracket sg \rrbracket \rightarrow g$. Any subset $C \subseteq \{1, \dots, n\}$ of $t + 1$ or more shares of the $\llbracket s \rrbracket$ can reconstruct the secret. Firstly, the recombination vector $\kappa = [\kappa_1, \dots, \kappa_n]$ is constructed by the generalized Lagrange basis polynomial that is accessible by all servers.

$$\kappa_i = \delta_i(0) = \prod_{j \in C, j \neq i} \frac{-j}{i-j}. \quad (3.2)$$

Therefore, the secret is reconstructed by polynomial,

$$s = f(0) = \sum_{i \in C} \kappa_i s_i \quad (3.3)$$

It can be extended for reconstructing the output of the SMPC protocols from the computed shares of the output by servers. Similarly, any set of $t + 1$ or more shares of the output $\llbracket sg \rrbracket$ can reconstruct the output. Figure 3-1 represents A schematic of the SMPC based on the secret sharing protocol for n parties $\mathcal{P}_1, \dots, \mathcal{P}_n$.

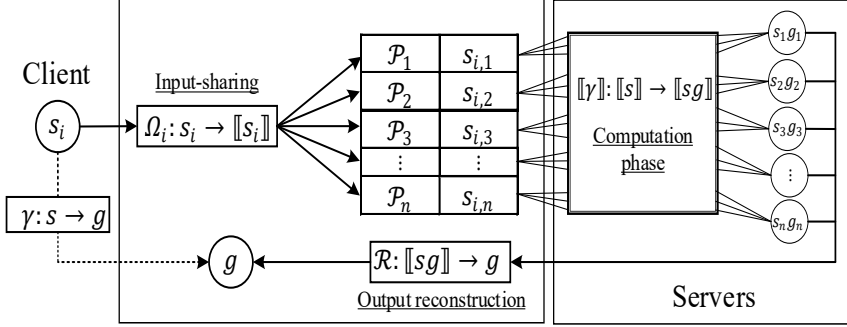


Figure 3-1 A schematic of the SMPC based on secret sharing scheme

As mentioned, the secret can be reconstructed by $t + 1$ or more of given shares via Lagrange interpolation theory. Furthermore, t of shares reveal nothing about the secret due to the uniformly random distribution of shares that implies the perfect privacy of Shamir's SSS. The secure addition protocol is simply achieved by doing the same operation on the shares, but the multiplication protocol needs more discussion and it limits the threshold of Shamir's SSS to $t < \frac{n}{2}$. It is assumed that servers have access to the multiplication protocol. Shamir's SSS allows a function of the secret to be computed directly on shares of secret. As observed, The SMPC protocols based on Shamir's SSS can be divided into three parts: Input-sharing, computation, and output-reconstruction phases that can be built as SMPC gates. Input-sharing and output-reconstruction are similar for all protocols based on Shamir's SSS. Hence, it is required to solely specify the computation phase in the SMPC protocols proposed in this paper. It is proven that all finite functions over a finite field can be represented as a polynomial over $\mathbb{F}_p$, which can be described using an arithmetic circuit. In this way, all that is needed in the computation phase are addition and multiplication.

CHAPTER 4. CONTROL STRATEGY

4.1. POLE PLACEMENT PROBLEM

A frequently used and very powerful method for designing state feedback gains for auto stabilization systems is the pole placement method. The task is to find a control law,

$$u(t) = -K^c x(t) \quad (4.1)$$

where $K^c_{r \times n}$ is the *state gain matrix*, and for the introduced system, the input u is of dimensions $r = 1$ and there are $n = 4$ state variables which then $K^c_{1 \times 4}$. The closed-loop state description becomes

$$\begin{cases} \dot{x} = (F - GK^c)x \\ y = Hx \end{cases} \quad (4.2)$$

This system describes an autonomous linear, time-invariant (LTI) system. If K^c can be chosen such that all eigenvalues of the $F - GK^c$ be in the open LHP, then $\lim_{t \rightarrow \infty} x(t) = 0$ and $\lim_{t \rightarrow \infty} y(t) = 0$. Therefore, this control law performs a regulation function and derives the state and the output to zero from any initial condition [28]. If regulation to some constant output set point $y_d \neq 0$ is requested, the dc steady-state must first be calculated for which y_d is the output. The steady-state input u^* and state x^* satisfy

$$\begin{cases} 0 = Fx^* + Gu^* \\ y_d = Hx^* \end{cases} \quad (4.3)$$

The new control law is $u = u^* - K^{cT} \Delta x$ where $\Delta x = x - x^*$. For the introduced system $x_1^* = x_3^* = \theta_{2d}$, $x_2^* = x_4^* = u^* = 0$ that the control law becomes

$$u(t) = -K_1^c(x_1 - x_1^*) - K_2^c x_2 - K_3^c(x_3 - x_3^*) - K_4^c x_4 \quad (4.4)$$

According to [23], to obtain a bandwidth of about 0.5 rad/s , the closed-loop control poles are selected at $-0.45 \pm 0.34j$ and $-0.15 \pm 1.05j$ which the desired characteristic polynomial is

$$p(s) = s^4 + 1.2s^3 + 1.7131s^2 + 1.10793s + 0.3578625 \quad (4.5)$$

Next, closed-loop characteristic polynomial is calculated by

$$p_c(s) = p_0(s) + K^{cT} \text{Adj}(SI - F)G \quad (2.6)$$

where $p_0(s) = \det(SI - F)$ is the open-loop characteristic polynomial.

$$p_c(s) = s^4 + (K_4^c + 0.0396)s^3 + (0.036K_4^c + K_3^c + 0.036K_2^c + 1.001)s^2 + (0.91K_4^c + 0.036K_3^c + 0.91K_2^c + 0.036K_1^c)s + (0.91K_3^c + 0.91K_1^c) \quad (4.7)$$

Regarding equations (4.5) and (4.7), the state gain matrix is obtained,

$$K^c = [-0.26139 \quad 0.041515 \quad 0.65538 \quad 1.1604] \quad (4.8)$$

4.2. ESTIMATOR (OBSERVER)

This design is based on full-state feedback. An estimator (observer) is needed to complete this optimal design. The full-order observer is of structure

$$\dot{\hat{x}} = F\hat{x} + Gu + K^e(y - H\hat{x}) \quad (4.9)$$

Here, $\hat{x} = [\hat{x}_1 \ \hat{x}_2 \ \hat{x}_3 \ \hat{x}_4]^T$ is the vector of estimated state variables that $\dim(\hat{x}) = n$ [same as $\dim(x)$] and K^e is a constant matrix $n \times m$ called the *observer gain*. The introduced system has $n = 4$ state variable and $m = 1$ output. This structure adds a correction term, $K^e(y - H\hat{x})$, to the previous observer. The equation of the estimation error is

$$\dot{\tilde{x}} = (F - K^e H)\tilde{x} \quad (4.10)$$

where $\tilde{x} = x - \hat{x}$. The system of equations (4.2) and (4.4) is observable if, and only if the dual system is controllable. if the system of equations (4.2) and (4.4) is observable, then (4.11) is controllable and a state gain matrix K^{eT} (an $m \times n$) matrix can be selected for arbitrary placement of the eigenvalues of the matrix $F^T x - H^T K^{eT}$.

$$\dot{x} = F^T x + H^T u \quad (4.11)$$

Hence, solving pole-placement for dual system (4.11) with desired error system poles leads to obtaining observer gain. The closed-loop error system poles are selected at $-7.7 \pm 3.12j$ and $-3.32 \pm 7.85j$ to be eight-time faster than the controller poles. In this way, a fast estimator has the same effect on the system response as no estimator at all which the desired characteristic polynomial is

$$p'(s) = s^4 + 22.04s^3 + 24.9253s^2 + 1577.0535s + 5014.27064 \quad (4.12)$$

And the closed-loop characteristic polynomial is

$$p'_c(s) = s^4 + (K_1^e + 0.0396)s^3 + (0.0396K_1^e + K_2^e + 1.001)s^2 + (0.091K_1^e + 0.0036K_2^e + 0.91K_3^e + 0.036K_4^e)s + (0.091K_2^e + 0.91K_4^e) \quad (4.13)$$

The observer gain is obtained

$$K^e = [22.0004 \quad 242.0531 \quad 1512.84032 \quad 5485.9822] \quad (4.14)$$

4.3. COMPENSATOR DESIGN

By combining the control law and estimator, the final compensator is formulated and the block diagram of such a system is illustrated in figure 4-1 where K^r is a scale factor to eliminate steady-state error which is also called *the feedforward gain*. In this way, the control law is obtained as

$$u(t) = K^r r(t) - K^c \hat{x}(t) \quad (4.15)$$

and K^r is calculated by

$$K^r = \frac{1}{-H(F - GK^c)^{-1}G} \quad (4.16)$$

For the nominal system under study, $K^r = 0.3933$.

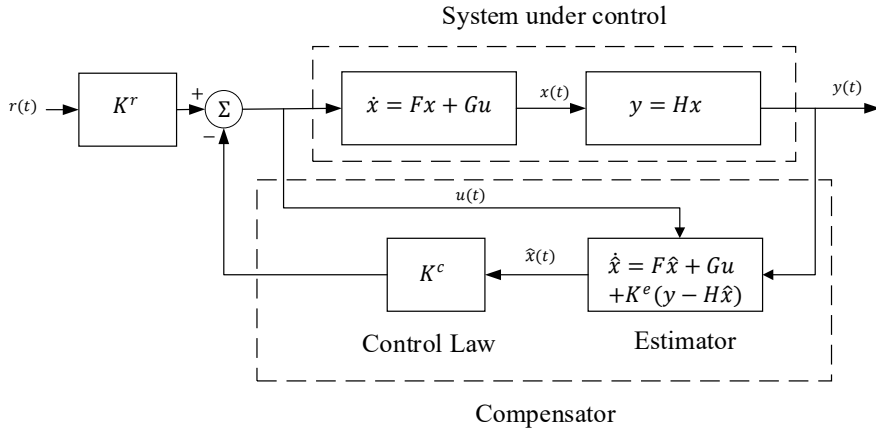


Figure 4-1 Block diagram of the system under control with compensator

The other way of describing the system is using the dc steady-state values and incremental control signal to obtain the control law that is formulated in (4.4). The block diagram of this representation is illustrated in figure 4-2.

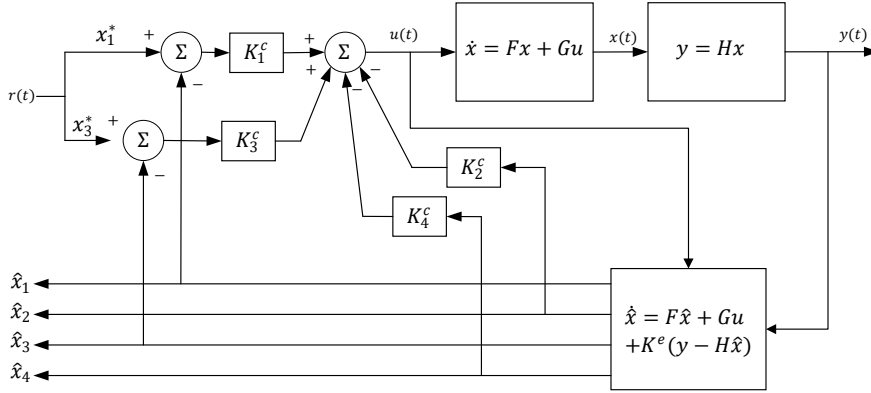


Figure 4-2 Block diagram of the system under control with compensator using incremental control signal

The compensator transfer function is calculated by,

$$Q_c(s) = \frac{U(s)}{Y(s)} = -K^c(sI - F + GK^c + K^eH)^{-1}K^e \quad (4.16)$$

that is,

$$Q_c(s) = \frac{-7361.7(s+0.318)(s^2+0.1858s+0.844)}{(s^2+15.4s+69.02)(s^2+6.64s+72.64)} \quad (4.17)$$

State-space representation of the closed-loop system can be written as,

$$\begin{cases} \dot{X} = \underbrace{\begin{bmatrix} F & -GK^c \\ -K^eH & F - K^eH - GK^c \end{bmatrix}}_{F_{cl}} X + \underbrace{\begin{bmatrix} G \\ G \end{bmatrix}}_{G_{cl}} K^r U \\ y = \underbrace{\begin{bmatrix} H & 0_{1 \times 4} \end{bmatrix}}_{H_{cl}} X \end{cases} \quad (4.18)$$

where $X = [x^T \hat{x}^T]^T$ is the state variable vector of the closed-loop system and $U = [u \ u]^T$. By substituting the system parameters,

$$F_{cl} = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ -0.91 & -0.036 & 0.91 & 0.036 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0.091 & 0.0036 & -0.091 & -0.0036 & 0.2614 & -0.04152 & -0.6554 & -1.16 \\ 22 & 0 & 0 & 0 & -22 & 1 & 0 & 0 \\ 242.1 & 0 & 0 & 0 & -243 & -0.036 & 0.91 & 0.036 \\ 1513 & 0 & 0 & 0 & -1513 & 0 & 0 & 1 \\ 5486 & 0 & 0 & 0 & -5486 & -0.03792 & -0.7464 & -1.164 \end{bmatrix}$$

$$G_{cl} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0.394 \\ 0 \\ 0 \\ 0 \\ 0.394 \end{bmatrix}, H_{cl} = [1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0]$$

The closed-loop transfer function is

$$Q_{cl}(s) = \frac{0.0142(s+25.28)}{(s^2+0.9s+0.3181)(s^2+0.3s+1.125)} \quad (4.19)$$

CHAPTER 5. PRIVACY PRESERVING COMPENSATOR PROTOCOL

In modern control systems, digital computers or microcontrollers are used in the real-time control of the physical system since they have the privileges of flexible control programming, decision making, and artificial intelligence which can be combined in dynamic control functions to achieve robust and multi-functional schemes [29]. In this section, a privacy-preserving SMPC protocol for secure estimation and control of the satellite is stated. But before that, the introduced compensator must be discretized to formulate the digital system implemented in the digital computer.

Digitization and sampling. Digital control system usually has a combination of both continuous-time and discrete-time systems, nevertheless, there may be some additional behavior in digital systems such as aliasing. The substantial differences between the two implementations are that the digital controller operates on samples rather than on continuous signals and the dynamics of the compensation are represented by algebraic recursive equations called the difference equations. Analog-to-digital (A/D) and digital-to-analog (D/A) converters are two Intermediary units between the physical and digital worlds. The former acts on physical variables mostly an electrical voltage measured by a sensor and converts it into a digital number. This unit is located before the control signal computation (by servers) and generates processable values via the controller. To the aim of keeping the sensor measurement private, data will be encrypted in the A/D converter (ADC) which changes its function to an enciphering unit rather than an encoder. As observed, the SMPC protocols take an integer number as input distributed by the input-provider while the system needs real number calculations that are real decision variables. Regarding this subject, the values are quantized with a suitable scale in the ADC unit to get the desired precision through the following transformation protocol,

Protocol 5-1: Enciphering (ADC) unit

Input: Analog signal $v^A \in \mathbb{R}$, truncate to $\sigma \in \mathbb{N}_0$ decimal place, and finite field $\mathbb{F}_p$.

Output: Shares of quantized $\llbracket v^D \rrbracket$.

1. Truncate v^A with given σ ,

$$\widetilde{v^A} = \text{trunc}(v^A, \sigma) = \begin{cases} \frac{\lfloor 10^\sigma v^A \rfloor}{10^\sigma} & v^A \geq 0 \\ \frac{\lceil 10^\sigma v^A \rceil}{10^\sigma} & \text{otherwise.} \end{cases} \quad (5.1)$$

2. Calculate and distribute scaled quantized parameter,

$$\llbracket v^D \rrbracket = \llbracket 10^\sigma \widetilde{v^A} \rrbracket \bmod p. \quad (5.2)$$

$\lfloor \cdot \rfloor$ and $\lceil \cdot \rceil$ denote the floor and ceiling operations respectively, $\widetilde{v^A}$ is the truncated signal, v^D is the integer that is shared by the dealer. It should be mentioned that all signals are discretized to be processable by a digital controller, while it will be observed that the values that are required to be private are encrypted. The act of the A/D unit leads to a quantization error that is an effect of digitization. The protocols also consider the quantization of multiplication because of extra scaling which is carried out securely by the servers. In this way, all calculations are performed with certain precision specified in the A/D converter. By contrast, the calculated shares of the control signal go through the D/A converter (DAC) to obtain the analog signal applied to the system which is considered a deciphering unit as the secret value is reconstructed in this stage. The protocol 5-2, represents the protocol of the DAC that outputs the real value $v^D \in \mathbb{R}$.

Protocol 5-2: Deciphering (DAC) unit

Input: The shares of digital value $\llbracket v^D \rrbracket$, $\sigma \in \mathbb{N}_0$, and finite field $\mathbb{F}_p$.

Output: Analog value v^A .

1. Reconstruct v^D , using *Output Reconstruction phase* of the Shamir's SSS.
2. For integer value v^D .

$$\begin{cases} v^A = \frac{v^D}{10^\sigma} & v^D \leq \frac{p-1}{2} \\ v^A = \frac{v^D-p}{10^\sigma} & \text{Otherwise.} \end{cases} \quad (5.3)$$

Indeed, the reconstructed field elements are compared in the DAC unit in order to evaluate $a < 0$ as $a \in \mathbb{F}_p$ with the following structure,

$$\underbrace{\left\{ 0, 1, \dots, \left\lfloor \frac{p}{2} \right\rfloor \right\}}_{\text{positive}}, \underbrace{\left\{ \left\lfloor \frac{p}{2} \right\rfloor + 1, \dots, p-1 \right\}}_{\text{negative}}$$

By this definition, the values are recognized to be signed integers or not. Therefore, the analog control signal can be reconstructed using protocol 5-2 before applying it to the system.

Digital Compensator. The differential equations describing the compensation with analog signal must be approximated by reducing them to algebraic equations representing the system with discrete data. It is assumed that all the numbers arrive with the same fixed sample period T . The continuous-time compensator system was designed for the plant in the prior section and converted to the discrete state-space model by the zero-order-hold method. The discrete-time state-space difference equations of the system with $T = 0.005$ sec, becomes,

$$\begin{cases} x(k+1) = Ax(k) + Bu(k) \\ y(k) = Cx(k) + Du(k) \end{cases} \quad (5.4)$$

$$A = \begin{bmatrix} 0.9998 & -0.004999 & 0 & 0 \\ 0.004999 & 1 & 0 & 0 \\ 1.25 \times 10^{-5} & 0.005 & 1 & 0 \\ 2.083 \times 10^{-5} & 1.25 \times 10^{-5} & 0.005 & 1 \end{bmatrix}, B = \begin{bmatrix} 0.004999 \\ 1.25 \times 10^{-5} \\ 2.083 \times 10^{-8} \\ 2.604 \times 10^{-11} \end{bmatrix} \quad (5.5)$$

$$, C = [0 \quad 0 \quad 0.036 \quad 0.9], D = 0$$

Where it takes the control parameter as input. Moreover, the discrete state-space model of the compensator is represented below, that inputs the sensor measurement and the calculated control signal $v = [u \ y]^T$.

$$\begin{cases} z(k+1) = A^c z(k) + B^c v(k) \\ u_c(k) = C^c z(k) + D^c v(k) \end{cases} \quad (5.6)$$

$$A^c = \begin{bmatrix} 1 & -1.27 \times 10^{-5} & -9.44 \times 10^{-4} & -0.02316 \\ 0.04 & 1 & -0.0024 & -0.05874 \\ 7.996 \times 10^{-4} & 0.03996 & 0.997 & -0.0733 \\ 2.075 \times 10^{-5} & 0.001542 & 0.07567 & 0.8928 \end{bmatrix}$$

$$, B^c = \begin{bmatrix} 1.303 \times 10^{-4} & -0.00664 \\ 1.813 \times 10^{-4} & -0.01312 \\ 1.282 \times 10^{-4} & -0.003186 \\ 9.078 \times 10^{-5} & 0.5447 \end{bmatrix} \quad (5.7)$$

$$, C^c = [0 \quad 0 \quad 0 \quad 64], D^c = 0$$

Privacy preserving challenges. The goal is to state privacy-preserving protocol that is satisfying the following objectives. The sensor measures the deviations from the star of the instrument package θ_2 or y that will be distributed among a small set of computing agents $\mathcal{P}_1, \dots, \mathcal{P}_n$. It is assumed that the network is interconnected through secure communication channels so that each server is able to send messages to another server. As mentioned, an input-provider or dealer is required to construct the Shamir's SSS polynomial and shares. Since there is just a system under control, sufficient computation power is considered in the system owning the secret (satellite) to invoke the protocol stated in the digitization part. In this way, no loss will be

entered into the real-world computation of SMPC protocols.

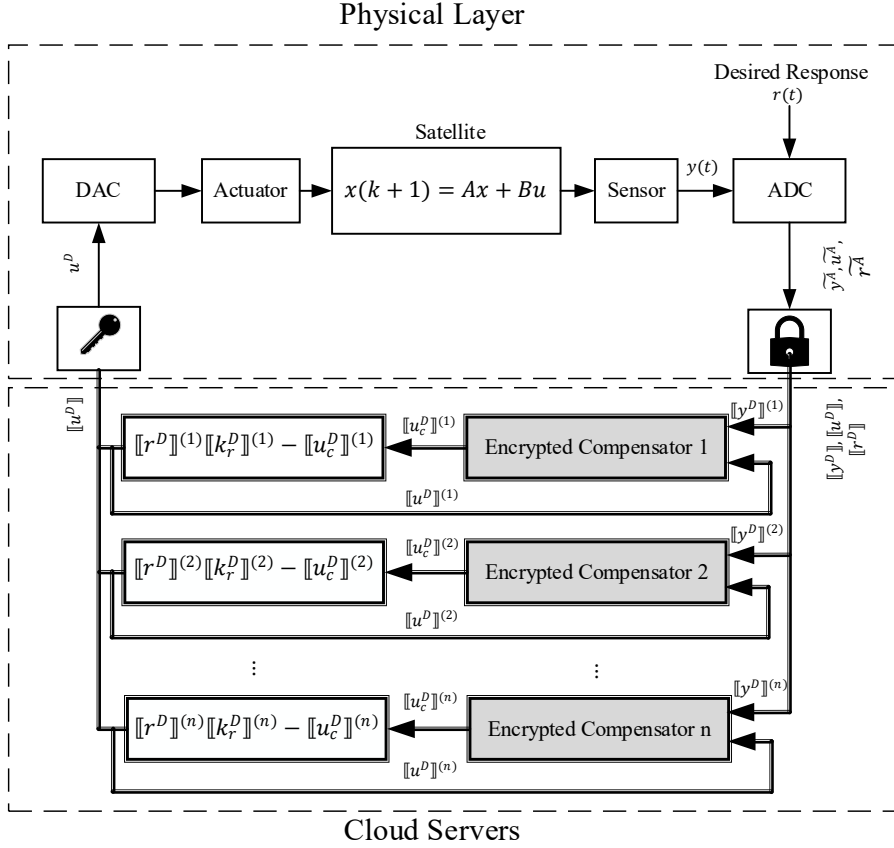


Figure 5-1 General block diagram of the system

A general block diagram of the system is observed in figure 5-1. Regarding this figure, the initial compensation states, the measured output of the system, the control signal, the desired response, and feedforward gain of the control system are shared between n agents and they execute the secure multiparty control algorithm and wish to control the system. The lock and key represent the input sharing and output reconstruction phases respectively. In addition, both estimations and the computing of the applied control signal to the system u^D is entirely are carried out by servers. The actuator is assumed to be ideal. The double framing shows the secure processing and the double arrows represent the secure communication channels between the servers enabling the bidirectional sending messages by servers to invoke the SMPC

algorithm. The secure algorithm of the proposed privacy-preserving protocol is stated in protocol 5-3.

Protocol 5-3: Distributed Privacy Preserving Satellite's Attitude Control

Initialization:

1. The parameters of the Shamir's SSS and digitization unit, finite field $\mathbb{F}_p$, number of the agents, privacy threshold t , and $\sigma \in \mathbb{N}_0$.
2. Agent $\mathcal{P}_i$ for $i = 1, \dots, n$ holds reference input $\llbracket r \rrbracket$, and feedforward gain $\llbracket k_r \rrbracket$.
3. Dealer initializes $u^0, y^0, x_1^0, x_2^0, x_3^0$, and x_4^0 .
4. Dealer initializes and distributes $\llbracket u^0 \rrbracket, \llbracket z_1^0 \rrbracket, \llbracket z_2^0 \rrbracket, \llbracket z_3^0 \rrbracket$, and $\llbracket z_4^0 \rrbracket$.
5. Determine the number of iterations with respect to runtime T_f and step time T as $N = \left\lfloor \frac{T_f}{T} \right\rfloor$.

for $k = 1, \dots, N$ **do**,

- Physical layer
 1. Reconstruct the control signal u^{k-1} from $k = 2$,
 2. $x_1^k = A_{1,1}x_1^{k-1} + A_{1,2}x_2^{k-1} + B_{1,1}u^{k-1}$,
 3. $x_2^k = A_{2,1}x_1^{k-1} + A_{2,2}x_2^{k-1} + B_{2,1}u^{k-1}$,
 4. $x_3^k = A_{3,1}x_1^{k-1} + A_{3,2}x_2^{k-1} + A_{3,3}x_3^{k-1} + B_{3,1}u^{k-1}$,
 5. $x_4^k = A_{4,1}x_1^{k-1} + A_{4,2}x_2^{k-1} + A_{4,3}x_3^{k-1} + A_{4,4}x_4^{k-1} + B_{4,1}u^{k-1}$,
 6. Sensor measures $y^k = C_{1,3}x_3^k + C_{1,4}x_4^k$,
 7. Dealer distribute $\llbracket y^k \rrbracket$ to agent $\mathcal{P}_i$ for $i = 1, \dots, n$.
 - Servo servers' computation of the compensation
 8. $\llbracket z_1^k \rrbracket = A_{1,1}^c \llbracket z_1^{k-1} \rrbracket + A_{1,2}^c \llbracket z_2^{k-1} \rrbracket + A_{1,3}^c \llbracket z_3^{k-1} \rrbracket + A_{1,4}^c \llbracket z_4^{k-1} \rrbracket + B_{1,1}^c \llbracket u^{k-1} \rrbracket + B_{1,2}^c \llbracket y^k \rrbracket$,
 9. $\llbracket z_2^k \rrbracket = A_{2,1}^c \llbracket z_1^{k-1} \rrbracket + A_{2,2}^c \llbracket z_2^{k-1} \rrbracket + A_{2,3}^c \llbracket z_3^{k-1} \rrbracket + A_{2,4}^c \llbracket z_4^{k-1} \rrbracket + B_{2,1}^c \llbracket u^{k-1} \rrbracket + B_{2,2}^c \llbracket y^k \rrbracket$,
 10. $\llbracket z_3^k \rrbracket = A_{3,1}^c \llbracket z_1^{k-1} \rrbracket + A_{3,2}^c \llbracket z_2^{k-1} \rrbracket + A_{3,3}^c \llbracket z_3^{k-1} \rrbracket + A_{3,4}^c \llbracket z_4^{k-1} \rrbracket + B_{3,1}^c \llbracket u^{k-1} \rrbracket + B_{3,2}^c \llbracket y^k \rrbracket$,
 11. $\llbracket z_4^k \rrbracket = A_{4,1}^c \llbracket z_1^{k-1} \rrbracket + A_{4,2}^c \llbracket z_2^{k-1} \rrbracket + A_{4,3}^c \llbracket z_3^{k-1} \rrbracket + A_{4,4}^c \llbracket z_4^{k-1} \rrbracket + B_{4,1}^c \llbracket u^{k-1} \rrbracket + B_{4,2}^c \llbracket y^k \rrbracket$,
 12. $\llbracket u_c^k \rrbracket = C_{4,1}^c \llbracket z_4^k \rrbracket$,
 13. Agents compute the control signal $\llbracket u^k \rrbracket = \llbracket r \rrbracket \llbracket k_r \rrbracket - \llbracket u_c^k \rrbracket$ and send it to dealer.
-

Theorem 1. (Correctness) The protocol 5-3 outputs the same result produced by the insecure control algorithm with a reasonable difference corresponding to the digitization part.

In fact, by increasing the value σ , the precision of the proposed SMPC protocol can be improved expectably which obviously goes along with a computation load. However, in order to capture the correct system dynamics, it is necessary that the accuracy considered for the digitization part is not lower than a certain value.

Theorem 2. (Privacy) The privacy of protocol 5-3 against the passive adversary follows from the fact that only secure stated protocols are utilized. This ensures the preservation of privacy up to the defined threshold of Shamir's SSS.

This theorem states that the privacy of the system is preserved while the adversary tries to learn the secret values through sabotage acts such as eavesdropping and spying.

CHAPTER 6. SIMULATION

In this section, simulations will be shown to indicate the performance of the proposed algorithm. For the simulation purpose, three servers are implemented by threshold $t = 1$ to observe the correctness of the algorithm. To obtain the closest result to the real case, the Gaussian noise is added to the sensor measurements which can be originated from communication channels and we compare the outputs with the insecure case when the same noises are applied. In this way, the robustness of the distributed privacy-preserving satellite's attitude control algorithm against noisy signals is discussed. Since the proposed SMPC compensation (SMPC-COM) algorithm is complex in terms of using secure operations, it is required to select the scale factor of the ADC unit as $\sigma = 7$. The transient-response specifications of the system are stated in table 6-1 and the *Integral Squared Error (ISE)* of the signals and the power of the signals are reported for the secure and insecure executions. The cardinality of the finite field is $\mathcal{P} = 18446744073709551629$. The standard deviation of the distribution for the noisy sensor is $\varepsilon = 0.005$ and the mean is $\mu = 0$.

Table 6-1 transient-response specifications of the system

Signal	Rise time (sec)	Overshoot (percent)	Settling Time (sec)
CL	2,4957	14,7817	15,6962
SMPC-COM	2,4957	14,7806	15,6944
Noisy CL	2,5127	16,2730	24,089
Noisy SMPC-COM	2,5127	16,2711	240,089

Figure 6-1 and 6-2 illustrate the closed-loop (CL) response of the system without and along with the noise respectively where the desired response is $r = 1$. The observations in the figures show the SMPC-COM algorithm produces the same results as the insecure control system.

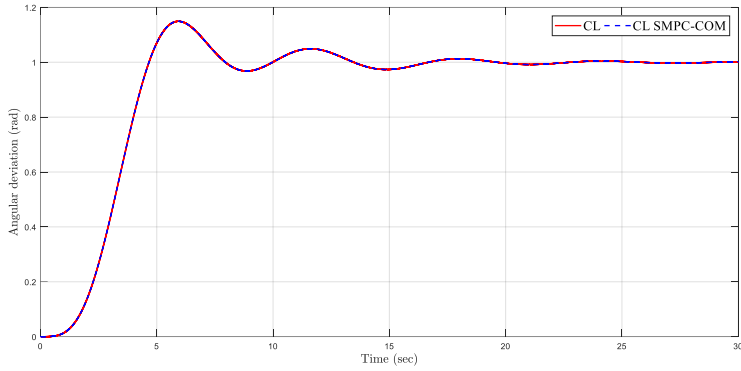


Figure 6-1 Closed-loop response of the system for insecure and SMPC-COM

Figure 6-3 illustrates the control signals in the mentioned states, in which both insecure and secure algorithms output similar responses. While with the noisy sensor, more effort is needed for control actions, the SMPC-COM algorithm is not sensitive to this which can also be seen in the comparison report in table 6-3. Figure 6-4 depicts the noisy sensor measurement and control signals where $\varepsilon = 0.001$.

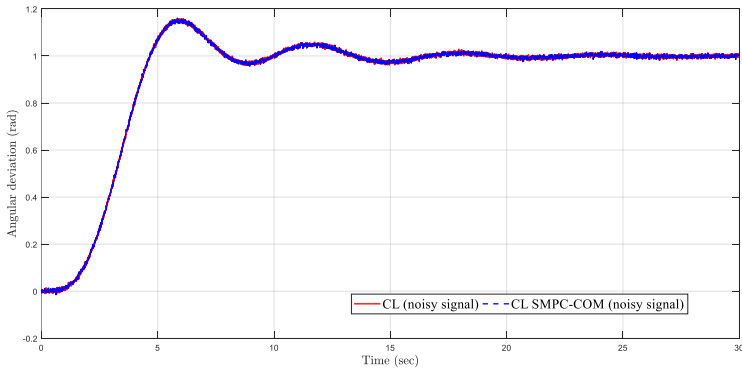


Figure 6-2 Closed-loop noisy response of the system for insecure and SMPC-COM algorithms

SIMULATION

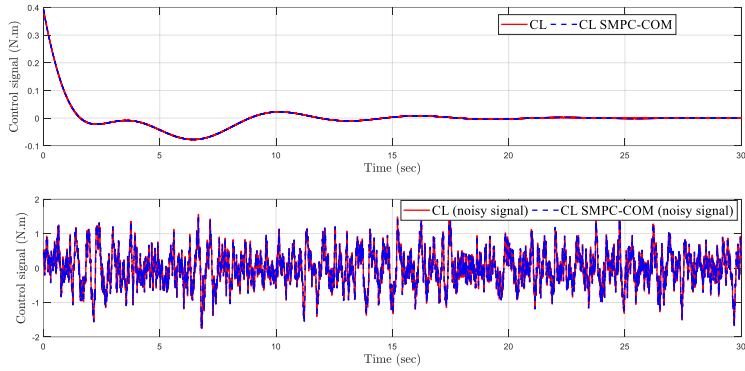


Figure 6-3 Control signal without and along with noise for insecure and SMPC-COM algorithms

Table 6-2 Power of signals and ISE of the system responses for insecure and SMPC-COM

Signal (without noise)	ISE	CL	CL SMPC-COM
x_1	$4,467 \times 10^{-10}$	$5,007 \times 10^{-3}$	$5,006 \times 10^{-3}$
x_2	$4,324 \times 10^{-10}$	$5,928 \times 10^{-3}$	$5,928 \times 10^{-3}$
x_3	$5,965 \times 10^{-10}$	$1,427 \times 10^{-2}$	$1,427 \times 10^{-2}$
x_4	$1,233 \times 10^{-9}$	1,111	1,111
z_1	$5,836 \times 10^{-12}$	$7,515 \times 10^{-3}$	$7,516 \times 10^{-3}$
z_2	$6,121 \times 10^{-10}$	$2,015 \times 10^{-1}$	$2,015 \times 10^{-1}$
z_3	$5,166 \times 10^{-8}$	46,47	46,47
z_4	$5,836 \times 10^{-12}$	$3,833 \times 10^{-5}$	$3,833 \times 10^{-5}$
Control signal u	$1,686 \times 10^{-9}$	$2,308 \times 10^{-3}$	$2,308 \times 10^{-3}$
θ_2	$1,001 \times 10^{-9}$	$9,015 \times 10^{-1}$	$9,015 \times 10^{-1}$

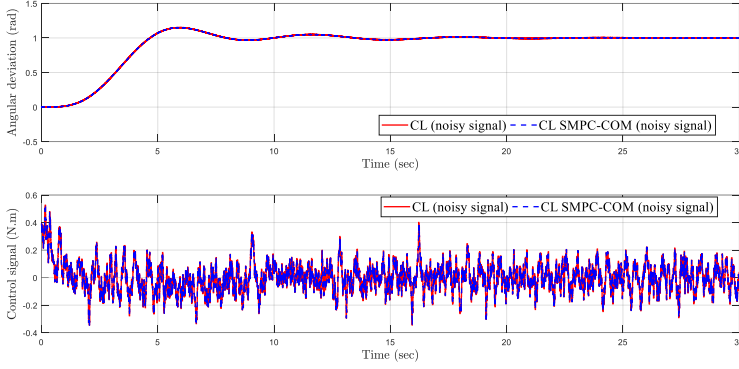


Figure 6-4 system response and control signal along with noise for insecure and SMPC-COM algorithms with $\varepsilon = 0.001$

Table 6-2 depicts signal power and ISE of the closed-loop (CL) system containing plant and compensator states, control signal, and sensor measurement for insecure CL and CL SMPC-COM while the noise is not present. The results of the noisy sensor are reported in table 6-3. Regarding the tables, the recorded outputs for both approaches have almost the same amounts. Figure 6-4 illustrates the ciphertext messages exchanged by the servers during the 50 iterations to compute the control signal $\llbracket u^k \rrbracket$ where the randomization and normal distribution of the messages can be observed.

Table 6-3 Power of signals and ISE along with noise of the system responses for insecure and SMPC-COM algorithms

Signal (without noise)	ISE	CL	CL SMPC-COM
x_1	$3,128 \times 10^{-10}$	$6,373 \times 10^{-3}$	$6,373 \times 10^{-3}$
x_2	$2,297 \times 10^{-10}$	$5,959 \times 10^{-3}$	$5,959 \times 10^{-3}$
x_3	$2,650 \times 10^{-10}$	$1,429 \times 10^{-2}$	$1,429 \times 10^{-2}$
x_4	$6,866 \times 10^{-10}$	1,111	1,111
z_1	$3,991 \times 10^{-12}$	$7,536 \times 10^{-3}$	$7,536 \times 10^{-3}$
z_2	$2,909 \times 10^{-10}$	$2,017 \times 10^{-1}$	$2,017 \times 10^{-1}$
z_3	$2,877 \times 10^{-8}$	46,47	46,47
z_4	$3,991 \times 10^{-12}$	$9,843 \times 10^{-5}$	$3,834 \times 10^{-5}$
Control signal u	$2,463 \times 10^{-9}$	$2,475 \times 10^{-1}$	$2,475 \times 10^{-1}$
θ_2	$5,567 \times 10^{-10}$	$9,015 \times 10^{-1}$	$9,015 \times 10^{-1}$

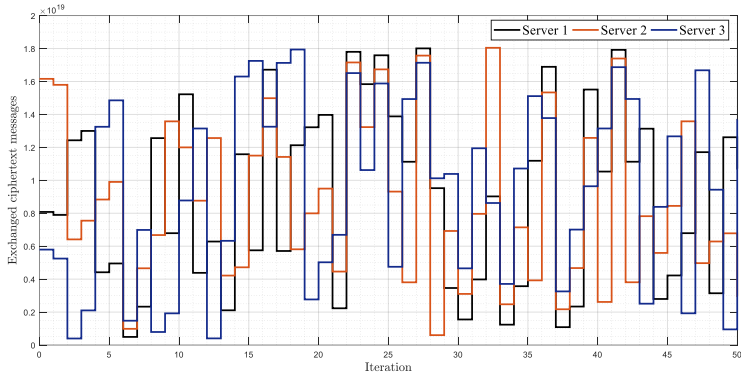


Figure 6-5 Exchanged ciphertexts by servers during the execution of SMPC-COM algorithm

CONCLUSION

In this work, a novel fully privacy-preserving digital distributed control algorithm based on SMPC secret sharing is proposed for pointing direction of a satellite while the sensor is mounted on the instrument package of the satellite. First of all, the dynamic of the system under control is extracted and analyzed including the equations of motion and modal analysis. An appropriate analog compensator is designed for the system using the pole-placement approach and discretized via the ZOH method in order to implement the digital algorithm.

Enciphering ADC and deciphering DAC units are introduced in the digitization part in order to communicate the finite field arithmetic of servers and the real decision variables of the physical world. The SMPC protocols are stated to secure the distributed digital compensator against the semi-honest adversary and any eavesdropping threat. The Computational servers collaborate through the stated protocol in each iteration and act on the shares which reveal nothing about the specified secret values up to the threshold of the privacy-preserving protocol.

Simulation results of the SMPC-COM algorithm, show the high efficiency and adequacy in comparison with the insecure implementation. The SMPC-COM algorithm produces the correct result in the noisy sensor case which can indicate the robustness of the proposed algorithm. The computation can be very expensive in terms of the communication as the number of operations is increased, nevertheless, this challenge can be mitigated through the use of a preprocessing phase.

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